Boolean SAT
Homework Assignment #1

The rules:

- The maximum number of points is stated for each problem; if a solution is partial, then only a part of these points will be granted.
- If a problem contains several items (a), (b), ..., they are graded separately.
- You need to get **40 points** in order to get the full grade for this assignment. Since the weight of this assignment is 40%, you can just think that you can earn up to 40 points towards the final grade (the final grade = up to 40 points for the first assignment + up to 40 points for the second assignment + up to 20 points for the final exam).

  For example, perfect solutions to problems 1.1(a,b) + 1.2(c,d) + 1.5 + 1.7(a) + 1.8 + 1.12 give you $8+16+4+5+4+3=40$ points, that is, the maximum. It is still reasonable to submit solutions for more problems than needed, just in case some of your solutions turn out to be imperfect.
- There are **12** problems (some of them are divided into parts/items).

Important dates:

- You can submit a *preliminary* solution of each problem to me by the end of January, by email to `edward.a.hirsch@gmail.com` . Please send in PDF, preferably one solution per file. You can submit a preliminary solution only once per problem/item. I will give you my comments on these solutions.
- The final solutions should be submitted using moodle by February 12 (inclusive). These solutions will be graded.

General note. You can use the results stated in the lectures without re-proving them. Note that the first three problems were given after the first two lectures, so you do not need anything else for solving them.

Problem HW1.1 (8 (upgraded!) points). Denote $F \overset{\mathbf{SAT}}{\equiv} G$ if $F \in \mathbf{SAT} \Leftrightarrow G \in \mathbf{SAT}$. A transformation of formulas $S(\,\cdot\,)$ is called satisfiability-equivalent if $\forall F\ F \overset{\mathbf{SAT}}{\equiv} S(F)$.

Prove that the following simplification rules are satisfiability-equivalent and show how to reconstruct a satisfying assignment for the original formula F from a satisfying assignment for the simplified formula:

(a) Blocked clause removal (the rule removes a blocked clause from F).

(b) $\mathrm{DP}_x(F)$ (the rule transforms F into $\mathrm{DP}_x(F)$).

The first item costs 3 points.
The second item costs 5 points (upgraded!).

Problem HW1.2 (38 (upgraded!) points). Consider the satisfiability problem for formulas that can be expressed as CSPs for relations of two types:

- clauses containing at most c variables,
- xor-equations containing at most d variables.

Using the DPLL approach, design an algorithm for this problem with a required running time bound (see below). Prove the correctness and the upper bound on the running time for the algorithm that you suggest.

(a) $c = 2$, $d = 6$, the exponent must be strictly better than ϕ^n.

(b) $c = 3$, $d = 15$, the exponent must be strictly better than $\tau^n_{(1,2,3)}$.

(c) $c = 2$, $d = 4$, the exponent must be strictly better than ϕ^n.

(d) $c = 3$, $d = 8$, the exponent must be strictly better than $\tau^n_{(1,2,3)}$.

While obviously $(a) \Rightarrow (c)$ and $(b) \Rightarrow (d)$, it is reasonable to try solving (c) and (d): they are much easier than (a) and (b).

Items (a) and (b) cost 11 points each (upgraded!).
Items (c) and (d) cost 8 points each (upgraded!).

Problem HW1.3 (20 points). Assume that we are given a formula in 3-CNF such that at least 1% of the complete assignments satisfy it. Find a satisfying assignment in polynomial time without the use of randomness.

Prove the correctness and the upper bound on the running time for the algorithm that you suggest.

Problem HW1.4 (5 points). Show that the derandomization of the PPZ algorithm for ***k*-SAT** presented in Lecture 3 takes at most $\widetilde{O}(2^{n(1-c_k)})$ time for $c_k = \frac{1}{2k} - o(\frac{1}{k})$, where the asymptotics in $o(\ldots)$ is as $k \to \infty$.

Problem HW1.5 (4 points). Suggest a min-2-wise independent small sample space (of course, with a proof) of size (much) less than the min-3-wise construction presented in Lecture 3.

Problem HW1.6 (5 points). Give a full proof for the min-3-wise independent derandomization construction for $k = 3$. In particular, handle the case $n - 1 \notin \mathbb{P}$, show that there is indeed exactly one invertible linear transformation that maps one triple into another one, and infer formally the claim about the probability (1/3).

Problem HW1.7 (10 points). In the clause shortening algorithm we used a ***k*-SAT** algorithm as a subroutine, it was done for $k = O(\log(m/n))$.

We estimated the clause shortening algorithm for the case where $m > n$.

(a) What can we say if $m \leqslant n$ — how much faster than 2^n can we solve **SAT** in this case?

(b) When we talked about PPZ, we assumed k to be a constant. However, we used this algorithm as a subroutine for a potentially non-constant $k = O(\log(m/n))$. What if $O(\log(m/n))$ is not a constant?

Each item (explaned rigorously with all the details!) costs 5 points.

Problem HW1.8 (4 points). Prove (rigorously!) that a composition of two SERF reductions is a SERF reduction.

Problem HW1.9 (15 (upgraded!) points). Write a detailed and rigorous proof of the general case in the theorem stating that $s_k \leqslant (1 - \Omega(\frac{1}{k}))s_\infty$: see the lecture notes for Lecture 4. You can (and will) use the bounds proven during the course of the uniquely satisfiable case without reproving them.

Problem HW1.10 (7 (upgraded!) points). Using the theorem that states that $s_k \leqslant (1 - \Omega(k^{-1}))s_\infty$ prove that s_k strictly increases ($s_{k+1} > s_k$) for an infinite number of the values of k. What hypothesis do you need for this? Why? (Explain by showing that otherwise some statement is wrong!)

Problem HW1.11 (10 points). Consider a generalized version of ***k*-SAT** where the usual clauses (disjunctions of literals) are generalized to arbitrary Boolean functions of k variables. (For example, $(x \oplus (y \wedge z))$, $((x \wedge \overline{z}) \vee (y \wedge z))$ and $(x \Rightarrow z)$ could be generalized clauses for $k = 3$.)

(Just a note: there are 2^{2^k} Boolean functions of k variables, and only 3^k of them are disjunctions of literals.)

How fast can you solve the satisfiability for such formulas? Provide an algorithm for this problem and prove an upper bound on its running time (do the best you can do based on the lectures). You can use randomness.

(a) $k = 3$,

(b) arbitrary k.

You need to prove your answer (everything that was not proved in the lectures needs to be proved).

Item **(a)** costs 3 points, item **(b)** costs 7 points.

Problem HW1.12 (3 points). Consider the resolution rule, that is, you can add a resolvent of any two clauses to the formula. Show using only this rule that the pigeon-hole formula (see the first lecture) for 3 pigeons and 2 holes is unsatisfiable. (No branching! You only add new clauses. Please write down the full sequence of new clauses, and from which old clauses they are derived.)

THE END